

Seyed Mohammad Ojagh Kazazi

M.Sc. Student in Electrical Engineering at EPFL

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Summary

Electrical engineer and M.Sc. student at EPFL working on computer vision, vision-language models, and efficient on-device AI. My work spans research, teaching, and hardware-aware systems, with a focus on models that remain accurate under real memory, latency, and reliability constraints.

Education

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| <p>M.Sc. École Polytechnique Fédérale de Lausanne (EPFL), Electrical Engineering</p> <ul style="list-style-type: none"> Fall 2026 coursework: Machine Learning, Mathematics of Data, Convex Optimization, Fundamentals of Inference and Learning. | <p>Lausanne, Switzerland
Sept 2026 – present</p> |
| <p>B.Sc. Sharif University of Technology, Electrical Engineering (Digital Systems)</p> <ul style="list-style-type: none"> GPA: 18.53/20 (3.95/4). Top 15% of the B.Sc. Electrical Engineering class and top 3 in Digital Systems. Selected graduate-level courses: Convex Optimization II (19.7), VLSI (19.1), Parallel Computing (18.4), Security in IoT (17.8), Deep Learning (17.2), Deep Generative Models (17.2). Selected undergraduate courses: Linear Algebra / Mathematical Methods (20), Data Structures and Algorithms (19.5), Communication Systems (19), Introduction to Machine Learning (17.9), ASIC/FPGA System Design (17.2). | <p>Tehran, Iran
Sept 2021 – 2025</p> |

Experience

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|---|---|
| <p>Aalto University, Research Assistant</p> <p>Supervisor: Prof. Siavash Khajavi.</p> <ul style="list-style-type: none"> Developed a knowledge-distillation framework combining KL-divergence and cross-entropy supervision to compress a 7B Qwen2.5 vision-language teacher into 0.5B–3B student models. On DetectiumFire, the 0.5B student retained 95.2% of the teacher’s burning-object accuracy with 14× fewer parameters and reached 99.5% fire/non-fire accuracy after deployment. Deployed and profiled distilled models on a Raspberry Pi 5 (4× Cortex-A76, 8 GB, CPU only): 1.9 GB peak memory, about 4 s end-to-end latency, and 2 false alarms on 1,452 test images. First-author submission to AAAI 2027 Innovative Applications of AI (IAAI). Currently investigating prompt optimization and reasoning strategies for Vision-Language-Action models. | <p>Espoo, Finland (remote)
Apr 2026 – present
6 months</p> |
| <p>Raderon AI Laboratory, Research Assistant</p> <p>Supervisor: Dr. S. Sarabi.</p> <ul style="list-style-type: none"> Conducted medical image analysis research combining the Discrete Wavelet Transform with swarm-based optimizers. Co-authored a paper published in Elsevier <i>Machine Learning with Applications</i>. | <p>British Columbia, Canada
(remote)
Aug 2024 – Jan 2025
6 months</p> |

Artificial Creature Laboratory (ACLab), Sharif University of Technology, Research Assistant

Supervisor: Prof. Saeed Bagheri Shouraki.

- Designed and built a smart IoT socket for the lab's custom AI platform, with a path toward commercialization.
- Evaluated IoT platforms and network protocols, then implemented firmware on Arduino and ESP32/ESP8266.

Tehran, Iran
July 2024 – Sept 2024
3 months

Teaching Experience

Sharif University of Technology, Laboratory Teaching Assistant

Logic Circuits and Digital Systems Lab, Fall 2023.

Tehran, Iran
Sept 2023 – Jan 2024

Sharif University of Technology, Teaching Assistant

Supported lectures, assignments, projects, and grading across electrical engineering and programming courses.

Tehran, Iran
Sept 2022 – Dec 2025

- Engineering Probability and Statistics — Fall 2025
- Logic Circuits and Digital Systems — four terms, 2023–2024
- Mathematical Methods in Engineering — Spring 2024
- Communication Systems — Spring 2024
- Object-Oriented Programming — Spring 2023
- Basic Programming — Fall 2022

Awards

National University Entrance Exam (Konkour)

Ranked in the top 0.1% of about 180,000 participants and admitted to Sharif University of Technology with a fully funded scholarship.

National Entrance Examination
2021

Academic Standing

Top 15% cumulative GPA among B.Sc. Electrical Engineering students of the 2021 class, including top 3 in Digital Systems.

Sharif University of Technology
2025

Publications

Distilling Vision-Language Models for On-Device Fire Understanding

Teacher–student distillation of domain-specialized VLMs for on-device fire understanding, with deployment on a commercial Detectium sensor.

Sept 2026

Mohammad Kazzazi, Zixuan Liu, Siavash Khajavi

[10.48550/arXiv.2609.05782](https://arxiv.org/abs/2609.05782) (arXiv preprint arXiv:2609.05782. Submitted to AAAI 2027 Innovative Applications of AI (IAAI))

CLEAR: A Closed-Form Minimal-Sensor TDOA/FDOA Estimator for Moving-Source IoT Localization

A closed-form two-stage TDOA/FDOA estimator for moving-source IoT localization with a minimal sensor set.

Oct 2025

Mohammad Kazzazi, Mohammad Morsali, Rouhollah Amiri

[10.48550/arXiv.2510.04160](https://arxiv.org/abs/2510.04160) (arXiv preprint arXiv:2510.04160. Submitted to IEEE Transactions on Aerospace and Electronic Systems)

Enhancing Skin Cancer Diagnosis Using Late Discrete Wavelet Transform and New Swarm-Based Optimizers

Mar 2026

Medical image analysis for skin cancer diagnosis using late discrete wavelet transform and swarm-based optimization.

Ramin Mousa, Saeed Chamani, Mohammad Morsali, **Mohammad Kazzazi**, Parsa Hatami, Soroush Sarabi

[10.1016/j.mlwa.2025.100811](https://doi.org/10.1016/j.mlwa.2025.100811) (Machine Learning with Applications (Elsevier))

Skills

Machine Learning and Computer Vision: PyTorch, Hugging Face Transformers, OpenCV, Ultralytics YOLO, scikit-learn, VLMs/VLAs, generative models, knowledge distillation, object detection, multi-object tracking, on-device inference

Programming: Python, C/C++, CUDA, Verilog, Java, MATLAB, Assembly

Hardware and Systems: Vivado, Xilinx ISE, ModelSim, LTspice, Zynq FPGA, Raspberry Pi, Arduino, ESP32/ESP8266

Tools: NumPy, pandas, Git, Linux, MySQL, LaTeX

Languages

Persian: Native

English: Fluent (TOEFL iBT 105)

Spanish: A1

Interests

Research: Computer vision, vision-language models, vision-language-action models, efficient and on-device deep learning, generative models, trustworthy machine learning, GPU computing, FPGA acceleration

Personal: Football, volleyball, gym, travel, sitcoms, classical Persian music, guitar

Projects

Football Object Detection and Multi-Object Tracking

Fine-tuned YOLOv8 on SportsMOT football video and built tracking with ByteTrack, DeepSORT, and CSRT, including occlusion and ID-switch analysis.

- mAP@50 0.947, mAP@50–95 0.779, precision 0.93, recall 0.94; 1.4 ms inference per frame on a P100 GPU.

Membership Inference Attacks Against Diffusion Models

Re-implemented SecMI (ICML 2023) on CIFAR-100 DDPMs and proposed a hyperparameter-free multi-timestep fusion variant.

- Raised attack AUC from 0.971 to 0.984 and TPR@1% FPR from 0.519 to 0.642.

Machine Unlearning and Private Training

Implemented SISA sharded training with ResNet-18 on CIFAR-10 and verified unlearning with backdoor success rate and membership inference.

Hardware and Parallel Computing

CORDIC and FFT on Zynq FPGA (Verilog, Vivado), a multi-cycle MIPS processor, parallel merge sort and heat-transfer simulation with texture memory in CUDA, and a digital telecommunication system simulation.

References

Prof. R. Amiri

Associate Professor, Electrical Engineering Department, Sharif University of Technology — amiri@sharif.edu

Prof. S. Bagheri Shouraki

Full Professor, Electrical Engineering Department, Sharif University of Technology — bagheri-s@sharif.edu

Prof. S. Khajavi Haghighat

Assistant Professor, Industrial Engineering and Management, Aalto University — siavash.khajavi@aalto.fi

Dr. S. Sarabi

Director, Raderon AI Lab — s.sarabi@raderonlab.ca